

Advanced (Carolina Reaper)

Q1. Simplify $a \cdot \sqrt{b}$ if $a = 2$ and $b = 8$

- (A) $4\sqrt{2}$
- (B) $2\sqrt{8}$
- (C) 4
- (D) $\sqrt{16}$
- (E) $\sqrt{32}$

Q2. What is the simplified form of $\sqrt{16} + \sqrt{9}$?

- (A) 5
- (B) 7
- (C) $4 + 3$
- (D) $\sqrt{25}$
- (E) 1

Q3. Simplify $(\sqrt{a} + \sqrt{b})^2$ if $a = 4$ and $b = 9$

- (A) $13 + 2\sqrt{36}$
- (B) $13 + 6\sqrt{2}$
- (C) 25
- (D) 36
- (E) 49

Q4. Simplify $\sqrt{a} - \sqrt{b}$ if $a = 16$ and $b = 4$

- (A) 2
- (B) 4
- (C) $\sqrt{12}$
- (D) $2\sqrt{3}$
- (E) $\sqrt{20}$

Q5. What is $\sqrt{12}$ simplified?

- (A) $2\sqrt{3}$
- (B) $\sqrt{3}$
- (C) $3\sqrt{2}$
- (D) $\sqrt{4}$
- (E) $\sqrt{6}$

Q6. What is the result of $\sqrt{20}$ simplified?

- (A) $2\sqrt{5}$
- (B) $\sqrt{5}$
- (C) $4\sqrt{5}$
- (D) $\sqrt{4}$
- (E) $\sqrt{10}$

Q7. Simplify $(\sqrt{a} \cdot \sqrt{b})^2$ if $a = 9$ and $b = 4$

- (A) 36
- (B) 6
- (C) 3
- (D) 4
- (E) 9

Q8. Simplify $\frac{\sqrt{a}}{\sqrt{b}}$ if $a = 16$ and $b = 9$

- (A) $\frac{4}{3}$
- (B) $\frac{3}{4}$
- (C) $\frac{2}{3}$
- (D) $\frac{16}{9}$
- (E) $\frac{9}{16}$

Open Ended Questions (FRQ)

Q9. Prove that $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$. Use $a = 9$ and $b = 4$ as an example.

Q10. Simplify the expression $(\sqrt{x} + \sqrt{y})^2$ and then find its value when $x = 16$ and $y = 9$. Show all your work.

Chef's Tips

- Emphasize the importance of simplifying radicals before performing operations.
- Make sure students understand that the denominator of a fraction must be rationalized when dealing with radicals.
- Encourage students to check their work by plugging in values for variables.